

2-Mark Questions & Answers — Data Structures Module 4: Trees

Q1. Define a Binary Tree and list its representations.

A **Binary Tree** is a hierarchical non-linear data structure where every node has at most two children (left and right).

- **Representations:**

1. **Sequential (Array-Based):** Left child at $2*i$, Right child at $2*i + 1$.
2. **Linked (Pointer-Based):** Nodes with `data`, `*left`, and `*right` pointers.

Q2. What is a Full Binary Tree vs Complete Binary Tree?

- **Full Binary Tree:** Every node has either 0 or 2 children.
- **Complete Binary Tree:** All levels are completely filled except possibly the last, which is filled from left to right without gaps.

Q3. Define Balance Factor in an AVL Tree.

The **Balance Factor (BF)** of a node in an AVL tree is defined as:

Balance Factor = Height(Left Subtree) - Height(Right Subtree)
Valid AVL Condition: $BF \in \{-1, 0, +1\}$

Q4. What is an Expression Tree?

An **Expression Tree** is a binary tree representation of an arithmetic expression where internal nodes are operators (`+`, `-`, `*`, `/`) and leaf nodes are operands (constants or variables).

Q5. Differentiate between B-Tree and B+ Tree.

- **B-Tree:** Stores keys and data record pointers in both internal and leaf nodes.
- **B+ Tree:** Stores data records only in leaf nodes; leaves are linked together sequentially for fast range queries.